$x = [x] + \{x\}$ then

$\uparrow$ integral $\quad \uparrow$ functional

① $[x+y] \geq [x] + [y]$

② $[x+n] \geq [x] + n$

③ $[x/n] = [[x]/n]$

$x - 1 < [x] \leq x$

and $0 \leq \{x\} < 1$

$\because$ $x = \xi n + \cancel{r} + \cancel{\alpha}$

$[x] = \xi n + \cancel{r}$

- $[n/m]$ counts $1, \dots, n$ divisible by m

- $\sum_{j=1}^{\infty} [n/p^j] =$ largest integer ℓ s.t. $p^\ell \mid n!$ $\leq \left[\frac{n}{p-1}\right]$

  $1 \dots n$ divisible by p^i

  geometric series

- $\binom{m}{n} = \frac{m!}{n!\,(m-n)!}$ integer $\quad \because$ $[m/p^j] \geq [n/p^j] + [(m-n)/p^j]$

  numerator $\quad$ denominator

- likewise $\frac{m!}{n_1! \cdots n_k!}$ integer

- multiplicative $f : \mathbb{N} \to \mathbb{R}$

① $(m,n) = 1$ then $f(mn) = f(m) f(n)$

② either $f(1) = 1$ or vanish identically

③ $f(n) = f(p_1^{j_1}) \cdots f(p_k^{j_k})$ if $n = p_1^{j_1} \cdots p_k^{j_k}$

④ $\sum_{d \mid n} f(d)$ is multiplicative $\quad \because$ $\sum_{d \mid m} f(d) \sum_{d' \mid n} f(d') = \sum_{d \mid m} \sum_{d' \mid n} f(d) f(d')$

$$= \sum_{dd' \mid mn} f(dd')$$

coprime $\quad$ multiplicative

Euler's (totient) func $\phi(n)$

---

① 분모가 n인 기약분수 $\phi(n) = n \prod_{p|n} (1-\frac{1}{p})$ "count 1, ..., n relatively prime to n"

② 분모가 n의 약수인 기약분수 $\sum_{d|n} \phi(d) = n$

$\frac{1}{6}$ $\frac{2}{6}$ $\frac{3}{6}$ $\frac{4}{6}$ $\frac{5}{6}$ $\frac{6}{6}$

$\phi(6)$ $\phi(3)$ $\phi(2)$ $\phi(3)$ $\phi(6)$ $\phi(1)$

$\phi(6) =$ (2분의 1) $\mu(1) \cdot 6$

(2의 배수) $\mu(2) \cdot 3$

(3의 배수) $\mu(3) \cdot 2$

(6의 배수) $\mu(6) \cdot 1 = 2$

- $n = p_1^{j_1} \cdots p_k^{j_k}$
- $\phi(n) = n \prod_{p|n} (1-\frac{1}{p})$

$$= n - \sum_r n/p_r + \sum_{r<s} n/(p_r p_s) - \sum_{r<s<t} n/(p_r p_s p_t) \cdots$$

n: 1 ... n

n/p_r: 1 .... n divisible by p_r

$n/(p_r p_s)$: 1 ... n divisible by $p_r p_s$

$n/(p_r p_s p_t)$: 1 ... n divisible by $p_r p_s p_t$

$$= \sum_{m=1}^{n} \underbrace{1 - \binom{l}{1} + \binom{l}{2} - \binom{l}{3} + \cdots}$$

$l(m)$: m을 나눌 수 있는 $p_1 \ldots p_k$

$$= \begin{cases} 1 & l = 0 \\ 0 & \text{otherwise} \end{cases}$$

i.e m/n은 기약분수

ϕ is multiplicative also by properties of congruences

- $\phi(p^j) = p^j - p^{j-1}$
- $\phi(1) + \phi(p) + \cdots + \phi(p^j) = 1 + (p-1) + \cdots + (p^j - p^{j-1}) = p^j$

$$\phi(n) = \sum_{m=1}^{n} \sum_{d|(m,n)} \mu(d) = \sum_{d|n} \mu(d) (n/d)$$

$d|(m,n)$: m/n은 기약분수?

$\mu(d)$: 제곱수 배제

$d|n$: 소수들의 곱

$\mu(d)$: 제곱수 배제

(n/d): d/n의 multiple

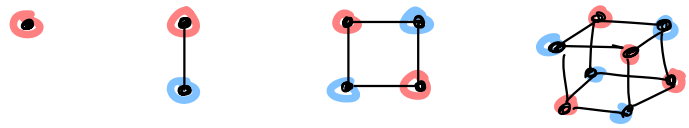
Möbius Function $\mu(n)$

- $\mu(n)$ and $\nu(n)$ multiplicative

- $\mu(n) = \begin{cases} 0 & \text{squared} \\ +1 & \text{even \# primes} \\ -1 & \text{odd \# primes} \end{cases}$

- $\nu(n) = \sum_{d|n} \mu(d) = \begin{cases} 1 & n=1 \\ 0 & n>1 \end{cases}$

$\nu(1)$ $\nu(2)$ $\nu(2\cdot 3)$ $\nu(2\cdot 3\cdot 5)$



$\Downarrow$ $\Downarrow$ $\Downarrow$ $\Downarrow$

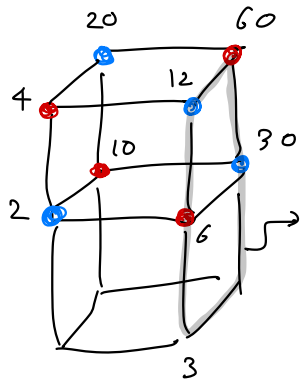
1 0 0 0

$\nu(n) = 0 \quad (n>1)$ because $\nu(p_i) = \underbrace{\mu(1)}_{+1} + \underbrace{\mu(p)}_{-1} + \underbrace{\mu(p^2)}_{\text{square}} + \cdots$

$= 0$

$$f(n) = g(n) - \sum_{p} g(n/p) + \sum_{p<q} g(n/pq) - \sum_{p<q<r} g(n/pqr)$$

$f(1)$	$f(2)$	$f(4)$
$f(3)$	$f(6)$	$f(12)$



$f(3)$은 $g(30)$, $g(60)$, $g(6)$, $g(12)$ 에 포함된다.

$$g(n) = \sum_{d|n} f(d) \iff f(n) = \sum_{d|n} \mu(d)\, g(n/d) = \sum_{d'|n} \mu(n/d')\, g(d')$$

Möbius Func은 순환사슬을 구한 Cech-Zolotarev

$(\Rightarrow)$ $f(n) = \sum_{d'|n} f(d')\, \nu(n/d')$ $\quad n = d'$

$= \sum_{d|n} \mu(d) \sum_{d'|n/d} f(d')$

$= \sum_{d|n} \mu(d) \cdot g(n/d)$

$(\Leftarrow)$ $g(n) = \sum_{d'|n} g(d')\, \nu(n/d')$ $\quad n = d'$

$= \sum_{d|n} \sum_{d'|n/d} \mu(n/dd')\, g(d')$

$= \sum_{d|n} f(n/d)$

$$g(x) = \sum_{n \le x} f(x/n)$$

$\therefore$ $\sum_{n\le x} \sum_{m \le x/n} \mu(n)\, f(x/nm)$

then $f(x) = \sum_{n \le x} \mu(n) \cdot g(x/n)$

$= \sum_{l \le x} f(x/l)\, \nu(l)$ $\quad l = 1$ $\quad (l = mn)$

5. $\tau(n)$, $\sigma(n)$

- # of divisors $\quad \tau(n) = \prod_{p \mid n} j+1 \qquad$ i.e $\sum_{d \mid n} 1$

- sum of divisors $\quad \sigma(n) = \prod_{p \mid n} (1+\cdots+p^j) \qquad$ i.e $\sum_{d \mid n} d$

$\Rightarrow$ τ, σ are multiplicative

---

① # of divisors

: $\tau(n) < c n^{\delta} \quad (\forall \delta > 0)$

- $f(n) = \tau(n) / n^{\delta}$ multiplicative

- $f(p^j) = \underbrace{(j+1)}_{\text{linear}} / \underbrace{p^{j\delta}}_{\text{exponential}} < 1$ except for $\underbrace{\text{finite } (j,p)\text{'s}}_{\text{select } c}$

---

② sum of divisors

: $\sigma(n) = \sum_{d \mid n} n/d \le \sum_{d \le n} n/d < n(1+\log n)$

---

③ 기약분수의 갯수

: $\phi(n) > \frac{1}{4} n / \log n \quad (n>1)$

- $f(n) = \dfrac{\sigma(n)\phi(n)}{n^2}$ multiplicative

- $f(p^j) = \dfrac{p^{j+1}-1}{p-1} \; p^{j-1}(p-1) / p^{2j}$

$= 1 - p^{-j-1} \ge 1 - 1/p^2$

- $\dfrac{\sigma(n)\phi(n)}{n^2} \ge \prod_{p \mid n} (1-1/p^2) \ge \prod_{m=2}^{\infty} (1-1/m^2) = 1/2$

6. Average Order of τ, σ, ϕ

---

① # of divisors

- $$\sum_{n \le x} \tau(\overset{\text{multiple}}{n}) = \sum_{d \le x} [x/\overset{\text{divisor}}{d}]$$

$$= x \log x + O(x) \qquad \because \sum_{d \le x} 1/d = \log x + O(1)$$

$$= x \log x + (2\gamma - 1) \qquad \text{(Euler Constant)}$$

$$+ O(\sqrt{x})$$

- average order : $\log n$ $\qquad \because \sum \log n \sim x \log x$

- $\underbrace{\text{almost all numbers}}$ have $(\log n)^{\log 2}$ divisors

i.e proportion $\le x$ without property is $o(x)$

i.e $\forall \varepsilon > 0$, almost all $n \in \mathbb{N}$

$$(\log n)^{-\varepsilon} < \tau(n) / (\log n)^{\log 2} < (\log n)^{\varepsilon}$$

---

② sum of divisors

- $$\sum_{n \le x} \sigma(n) = \sum_{d \le x} \sum_{m \le x/d} m \qquad (n = dm)$$

$$= \sum_{d \le x} \frac{1}{2} [x/d]([x/d]+1)$$

$$= \sum_{d \le x} \frac{1}{2} (x/d)^2 + O(x/d)$$

$$= \frac{1}{12} \pi^2 x^2 + O(x \log x) \qquad \because \sum_{d \le x} 1/d^2 = \sum_{d=1}^{\infty} 1/d^2 + O(1/x)$$

- average order : $\frac{1}{6} \pi^2 n$ $\qquad \because \sum n \sim \frac{1}{2} x^2$

③ 기약분수의 갯수

- $\sum_{n \le x} \phi(n) = \sum_{n \le x} \sum_{d \mid n} \mu(d) \cdot (n/d)$

$$= \sum_{d \le x} \mu(d) \cdot \sum_{m \le x/d} m$$

$$= \sum_{d \le x} \mu(d) \left(\frac{1}{2} (x/d)^2 + O(x/d) \right)$$

$$= (3/\pi^2)\, x^2 + O(x \log x)$$

$\because \sum_{d \le x} \mu(d)/d^2 = \sum_{d=1}^{\infty} \mu(d)/d^2 + O(1/x)$

- average order : $6n/\pi^2$

$Pr(\text{coprime}) = 6/\pi^2 \approx 60\%$

$\because$ $1 \le p \le q \le n$ then # of pairs : $n(n+1)/2$

# of pairs : $\phi(1) + \cdots + \phi(n)$

(lowest term)

7. Perfect Number

- average order of $\sigma(n)$: $1.64n$

n is perfect means $\sigma(n) = 2n$

- $\sigma(6) = 1+2+3+6$ i.e $6 = 2^1 \underbrace{(2^2-1)}_{=3}$
- $\sigma(28) = (1+2+4)(1+7) = 56$ i.e $28 = 2^2 \underbrace{(2^3-1)}_{=7}$

does odd perfect number exists ?

even number is perfect $\Longleftrightarrow$ $2^{p-1}(2^p-1)$ with p, 2^p-1 prime

- perfect $n = 2^k \cdot \underbrace{m}_{\text{odd}}$
- (perfect) $\sigma(n) = 2n$
- (multiplicative) $\sigma(n) = \sigma(2^k) \cdot \sigma(m)$
- $\sigma(m) = \dfrac{2n}{\sigma(2^k)} = \dfrac{2 \cdot 2^k \cdot m}{\underbrace{2^{k+1}-1}_{=l}} = 2^{k+1} \cdot l$

$$= (2^{k+1}-1)\, l + l$$

$$= m + l$$

- therefore $l = 1$

$\Rightarrow$ $m = 2^{k+1} - 1$ is Mersenne prime

$\Rightarrow$ $k+1$ is prime.

even perfect $\xleftrightarrow{\text{bijection}}$ Mersenne prime

8. The Riemann zeta-function

- prime distribution

  related with zeta function $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s \quad (s = \sigma + it)$

Note. $\dfrac{1}{n^s} = \underbrace{\dfrac{1}{n^\sigma}}_{\text{크기}} \cdot \underbrace{\dfrac{1}{n^{it}}}_{\text{방향}}$ with $\left|\dfrac{1}{n^s}\right| = \dfrac{1}{n^\sigma}$

- $n^{it} = e^{it\log n} = \cos(t\log n) + i\sin(t\log n)$
- therefore $|n^{it}| = 1$

- $\sigma > 1$ then $\zeta(s)$ converges absolutely $\because \left|\dfrac{1}{n^s}\right| = \dfrac{1}{n^\sigma}$
- $\sigma > 1+\delta$ then $\zeta(s)$ converges uniformly $\because \dfrac{1}{n^\sigma} \le \dfrac{1}{n^{1+\delta}}$ (M-test)

- $\zeta(s)$ continued analytically throughout complex plane
- regular except for simple pole at $s=1$ with residue 1.

  i.e $\zeta(s) = \boxed{\dfrac{1}{s-1}} + c_0 + c_1(s-1) + c_2(s-1)^2 + \cdots$

- $\Xi(s) = \pi^{-\frac{1}{2}s}\,\Gamma(\tfrac{1}{2}s)\,\zeta(s)$ Note. $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$
- $\Xi(s) = \Xi(1-s)$ $= (z-1)!$

---

claim $\zeta(s) = \dfrac{\pi^{s/2}\cdot\Xi(1-s)}{\Gamma(s/2)} = 0$ at $s = -2n$

$\because$ (해석적 연속) $\Gamma(z) = \Gamma(z+1)/z$

(무한대로 발산) $\Gamma(-n) = \dfrac{\Gamma(1)}{(-n)(-n+1)\cdots(-1)\,0}$

$$\sum_{n=1}^{\infty} 1/n^s = \prod_p (1-1/p^s)^{-1} \qquad (\sigma > 1)$$

zeta function Euler product

$\because$ • $\prod_{p \le N} (1-1/p^s)^{-1} = \prod_{p \le N} (1+p^{-s}+p^{-2s}+\cdots)$ (m : divisible only by primes $\le N$)

$$= \sum_m m^{-s}$$

• $\left| \sum_m m^{-s} - \sum_{n \le N} n^{-s} \right| \le \sum_{n > N} n^{-\sigma} \longrightarrow 0$ (as $N \to \infty$)

• $\sigma > 1$ then $\zeta(s)$ no zeros

• $\sigma < 0$ then $\zeta(s)$ only trivial zeros at $-2, -4, -6 \cdots$

• $0 \le \sigma \le 1$ then $\zeta(s)$ has

$\underbrace{\text{infinitely many}}_{\text{Hardy}}$ zeros $\underbrace{(\text{only?})}_{\text{Riemann}}$ in $\sigma = \frac{1}{2}$

---

if Riemann Hypothesis true

① $\pi(x) = \int_2^x \frac{dt}{\log t} + O(\sqrt{x} \log x)$ and $p_{n+1} - p_n = O(p_n^{\frac{1}{2}+\varepsilon})$

Note. this is valid in weaker term

with narrow zero-free region left of line $\sigma = 1$

---

② $\sum_{n \le x} \mu(n) = O(x^{\frac{1}{2}+\varepsilon})$ because $1/\zeta(s) = \sum_{n=1}^{\infty} \mu(n)/n^s$

• $\sigma > 1$ then $\zeta(s) \cdot (1/\zeta(s)) = \sum_{n=1}^{\infty} \frac{1}{n^s} \nu(n)$

• vaild in $\sigma > \frac{1}{2}$ with Riemann Hypothesis

---

③ $\sigma > 2$ then $\zeta(s-1)/\zeta(s) = \sum_{n=1}^{\infty} \phi(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{n}{n^s}\right)\left(\sum_{d=1}^{\infty} \frac{\mu(d)}{d^s}\right)$

④ $\sigma > 1$ then $(\zeta(s))^2 = \sum_{n=1}^{\infty} \tau(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \cdot \left(\sum_{d=1}^{\infty} \frac{1}{d^s}\right)$

⑤ $\sigma > 2$ then $\zeta(s)\zeta(s-1) = \sum_{n=1}^{\infty} \sigma(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \cdot \left(\sum_{n=1}^{\infty} \frac{d}{d^s}\right)$